Q TL-seq GUI

Ver. 2.0

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introduction

QTL-seq GUI

QTL-seq GUI is a tool for rapidly identifying genomic regions associated with specific phenotypes in populations derived from crosses between different varieties, using whole-genome sequencing data.
For issue reports or requests regarding this tool, please contact us via "Issues" or "Pull Request" on GitHub (https://github.com/sakemiki/QTL-seq_GUI_ver2).

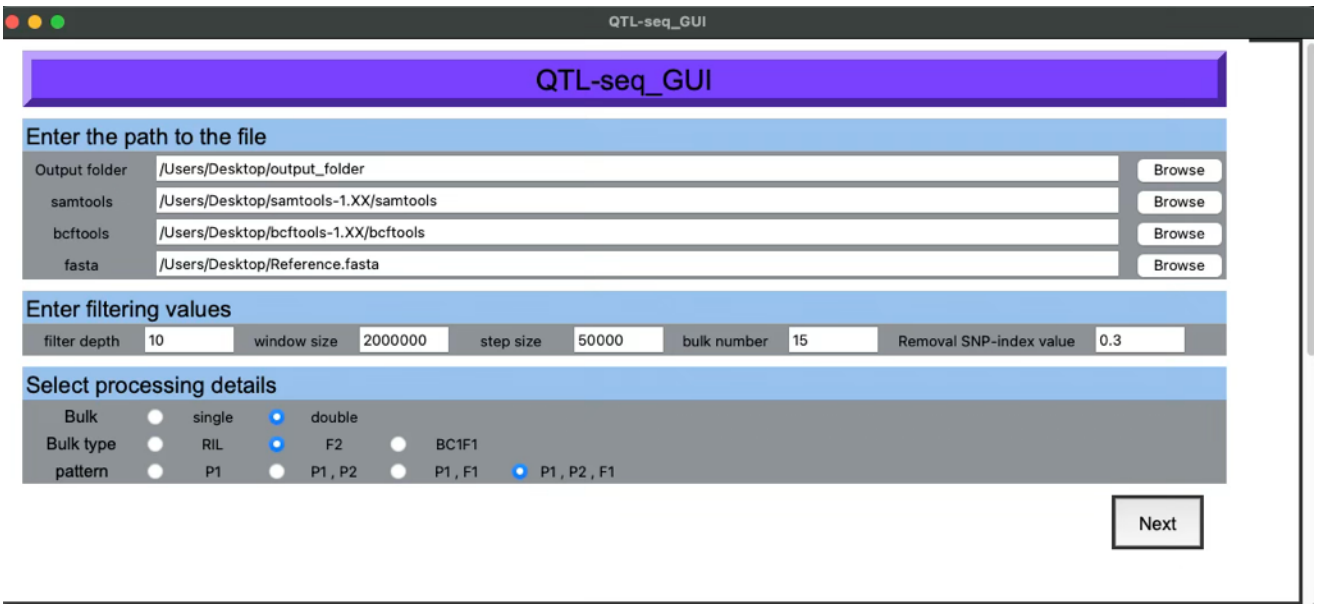
preparation

Install the following tools:
SAMtools (URL : http://www.htslib.org/download/)
BCFtools (URL : http://www.htslib.org/download/)
Python3 (URL : https://www.python.org/downloads/)

FASTA file : Reference genome file
BAM files: sorted alignment files (one per sample)

⚠Ensure that a corresponding .bam.bai index file exists in the same directory as each BAM file. If the index file is missing, the process will abort.

Using the GUI



Files and folders can be specified in the following three ways:

• Write the Full Path: You can directly input the full path of the file or folder in the input field.

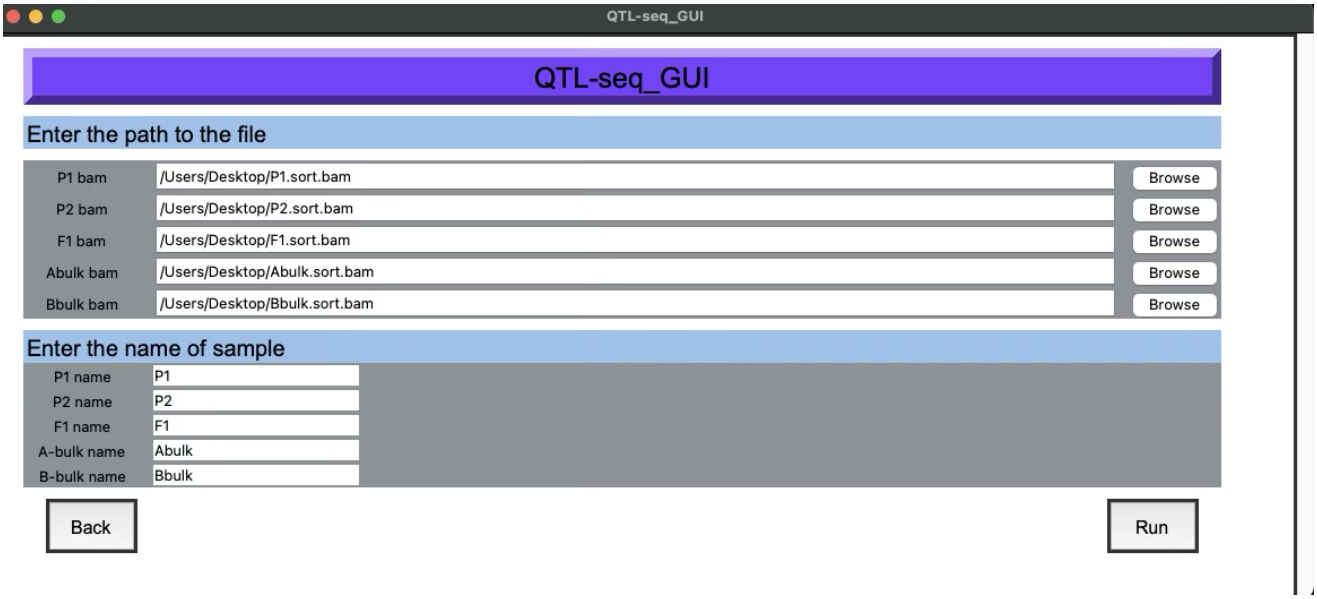
• Use the Browse Button: You can click on the "Browse" button, which will open a file dialog.

• Drag and Drop: You can select the file or folder from your file explorer and drag it to the input field in the QTL-seq GUI.

Next:

After entering the output folder, tool paths, and FASTA file path, along with the required filtering options, click the 'Next' button in the lower right corner.

Using the GUI



Run:

After selecting the bulk type and filter pattern, enter the paths to the required BAM files and the corresponding sample names, then click the '**Run**' button in the lower right corner to start the analysis.

Back:

To return to the previous screen, click the '**Back**' button.

Using the GUI

The QTL-seq_GUI has the following three input fields.

1. Enter the path to the file
Drag and drop or directly enter the path to the output folder, required tools, and the BAM file to be used, or select using the "Browse" button.

2. Enter BAM file names and filtering values
Enter the names of the BAM files to be used and the numeric values for the filtering parameters.

3. Select processing details
Please select the appropriate option. 'Pattern' specifies which type of BAM files are used for filtering.

Run

1. Enter the path to the file：

Output Folder:

Select the folder where QTL-seq GUI output will be saved.

⚠Caution: Selecting a previously used output folder may overwrite existing results.

SAMtools

Enter the path to the SAMtools executable.

BCFtools

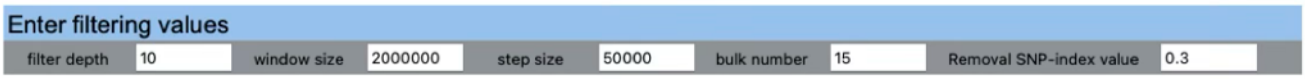
Enter the path to the BCFtools executable.

fasta

Enter the path to the FASTA file.

Run

1. Enter filtering values：



filter depth
Enter the numerical value of the depth to be used for filtering.

window size
Sliding window size (bp): 2000000

step size
Sliding window step size (bp): 50000

bulk number
Enter the number of individuals used to construct each bulk. If the two bulks contain different numbers of individuals, enter the smaller value.

Removal SNP-index value
Positions in both bulk datasets with a SNP-index below this value are excluded, to filter out false positive SNPs.

Run

1. Select processing details：

Bulk
Select '**Single**' if only one bulk is being analyzed, or '**Double**' if two bulks are being analyzed.

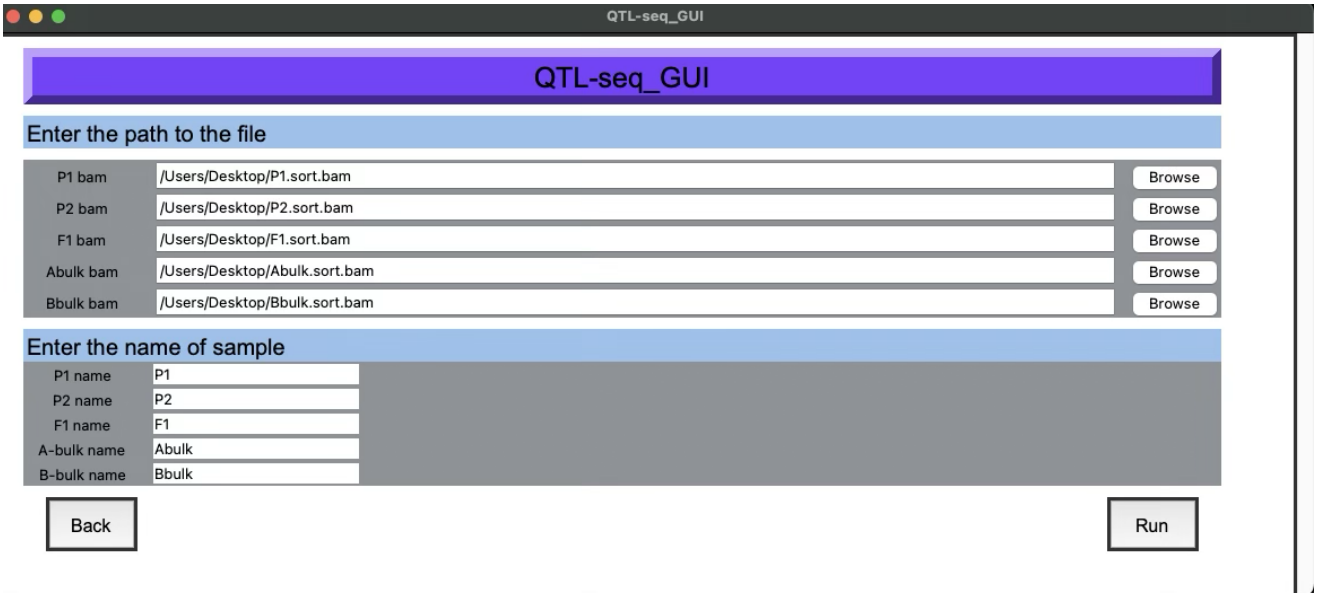
Bulk type
Select the bulk type that matches your experimental population: **RIL**, **F2**, or **BC1F1**.

pattern
Select the appropriate filter pattern based on which BAM files are available for your analysis.

⚠ Each sample must be assigned a unique name. Duplicate sample names will cause an error.

Run

2. Enter the path to the file



Input fields will appear for each BAM file required for the analysis. Fill in all fields before proceeding.
Then, click the "**Run**" button in the lower right corner to execute the process.
If you wish to return to the previous screen, please click the "**Back**" button in the lower left corner.

Output

QTL-seq_GUI generates three types of folders.

```
<Output folder>
    | ---bam
    | ---txt
    | ---result
```

The files output to each folder are described on the following pages.

Output

Example of output file when the following contents are selected

```
bam
 |---P1_chrxxx.bam
 |---P1_chrxxx.bam.bai
 |---P2_chrxxx.bam
 |---P2_chrxxx.bam.bai
 |---F1_chrxxx.bam
 |---F1_chrxxx.bam.bai
 |---Abulk_chrxxx.bam
 |---Abulk_chrxxx.bam.bai
 |---Bbulk_chrxxx.bam
 |---Bbulk_chrxxx.bam.bai      etc
```

BAM files and their corresponding BAM index files (.bam.bai) are output, split by chromosome.

Output

Example of output file when the following contents are selected

```
txt
 |———-QTLseq_input_log.txt
 |———-snpindex_merge.txt
 |---average_snp_index_per_window.txt
 |---confidence_interval.txt
 |---Abulk-Bbulk_95_peak.txt
 |---Abulk-Bbulk_99_peak.txt
```

txt

QTLseq_input_log.txt

```
=== QTL-seq run parameters ===
Output folder: /Users/Desktop/output_folder/txt
samtools: /Users/Desktop/samtools-1.20/samtools
bcftools: /Users/Desktop/bcftools-1.20/bcftools
FASTA: /Users/Desktop/Reference.fasta
P1 BAM: /Users/Desktop/P1.sort.bam (P1)
P2 BAM: /Users/Desktop/P2.sort.bam (P2)
F1 BAM: /Users/Desktop/F1.sort.bam (F1)
Abulk BAM: /Volumes/Abulk.sort.bam (Abulk)
Bbulk BAM: /Volumes/Bbulk.sort.bam (Bbulk)
Filter Depth: 10
Window Size: 2000000
Step Size: 50000
Bulk Number: 15
Removal SNP-index threshold: 0.3
Bulk: Double
Bulk Type: F2
Filter Pattern: P1 , P2 , F1
```

① Output folder: /Users/Desktop/output_folder/txt … Bbulk BAM: /Volumes/Bbulk.sort.bam (Bbulk)

② Filter Depth: 10 … Removal SNP-index threshold: 0.3

③ Bulk: Double … Filter Pattern: P1 , P2 , F1

The information listed here is what you entered on the input screen.

① is the path to the file you entered. The text in parentheses is the sample name.

② is the numerical value of the filter used for analysis.

③ is which filter process was used.

txt

snpindex_merge.txt

This file contains SNPs extracted from the VCF file that meet the following criteria: sequencing depth ≥ the set threshold, homozygous in P1 and P2, and heterozygous in F1. Chromosome name, position, SNP-index and depth information for each sample are listed in tab-delimited format.

①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪	⑫	⑬
chr1	3500	0	25	1	25	0.556	18	0.333	21	0.483	29	-0.15
chr1	4500	0	19	1	19	0.471	17	0.571	14	0.429	21	0.142
chr2	7500	1	22	0	22	0.52	25	0.368	19	0.565	23	-0.197
chr2	8500	0	19	1	19	0.5	16	0.333	15	0.529	17	-0.196
chr3	10500	0	14	1	14	0.526	19	0.48	25	0.235	17	0.245
chr3	11500	0	17	1	17	0.35	20	0.25	12	0.452	31	-0.202
chr4	12500	0	22	1	22	0.5	18	0.556	18	0.471	17	0.085
chr4	15500	0	17	1	17	0.652	23	0.5	22	0.4	20	0.1

①; chr
②; position
③; P1 snp-index
④; P1 depth
⑤; P2 snp-index
⑥; P2 depth
⑦; F1 snp-index
⑧; F1 depth
⑨; Abulk snp-index
⑩; Abulk depth
⑪; Bbulk snp-index
⑫; Bbulk depth
⑬; delta snp-index

txt

average_snp_index_per_window.txt
This file contains the average SNP-index for each bulk and the delta SNP-index, calculated within each sliding window. Each row includes the chromosome name, window midpoint, start position, end position, and average SNP-index values, in tab-delimited format.

①	②	③	④	⑤	⑥	⑦
chr1	1000000	0	2000000	0.53	0.43	0.096
chr1	1050000	50000	2050000	0.58	0.45	0.125
chr2	1000000	0	2000000	0.77	0.36	0.416
chr2	1050000	50000	2050000	0.76	0.32	0.444
chr3	1000000	0	2000000	0.5	0.43	0.071
chr3	1050000	50000	2050000	0.53	0.39	0.136

①; chr
②; middle position
③; start position
④; end position
⑤; Abulk SNP-index
⑥; Bbulk SNP-index
⑦; delta SNP-index

txt

confidence_interval.txt

This file contains the confidence intervals calculated from depth-based simulations. Each row includes the chromosome name, window midpoint, start position, end position, and the 95% and 99% confidence intervals for both the SNP-index and delta SNP-index, in tab-delimited format.

①	②	③	④	⑤	⑥	⑦	⑧	⑨	⑩	⑪	⑫
chr1	1000000	0	2000000	0.15	0.86	0.225	0.78	-0.511	0.52	-0.386	0.396
chr1	1050000	50000	2050000	0.15	0.85	0.226	0.78	-0.507	0.51	-0.379	0.392
chr2	1000000	0	2000000	0.15	0.86	0.228	0.78	-0.51	0.51	-0.389	0.4
chr2	1050000	50000	2050000	0.15	0.87	0.227	0.79	-0.516	0.52	-0.394	0.41
chr3	1000000	0	2000000	0.15	0.86	0.221	0.79	-0.512	0.52	-0.391	0.402
chr3	1050000	50000	2050000	0.14	0.86	0.22	0.79	-0.517	0.53	-0.389	0.402

①; chr
②; middle position
③; start position
④; end position
⑤; Lower mean snp-index at 99% confidence interval
⑥; Higher mean snp-index at 99% confidence interval
⑦; Lower mean snp-index at 95% confidence interval
⑧; Higher mean snp-index at 95% confidence interval
⑨; Lower value of delta mean snp-index at 99% confidence interval
⑩; Higher value of delta mean snp-index at 99% confidence interval
⑪; Lower value of delta mean snp-index at 95% confidence interval
⑫; Higher value of delta mean snp-index at 95% confidence interval

txt

Abulk-Bbulk_95_peak.txt

This file lists genomic regions where the delta SNP-index exceeds the 95% confidence interval, indicating candidate QTL peaks.

Chromosome name, start position, end position, and peak position are described in tab-delimited format.

①	②	③	④	⑤	⑥	⑦
chr2	0	2050000	1050000	0.76	0.32	0.444
chr03	0	5000000	1300000	0.866	0.183	0.684
chr03	22200000	25000000	23500000	0.858	0.344	0.515
chr08	0	5100000	1800000	0.877	0.252	0.625

①; chr
②; start position
③; end position
④; peak position
⑤; Abulk SNP-index
⑥; Bbulk SNP-index
⑦; delta SNP-index

txt

Abulk-Bbulk_99_peak.txt

This file lists genomic regions where the delta SNP-index exceeds the 99% confidence interval, indicating candidate QTL peaks.

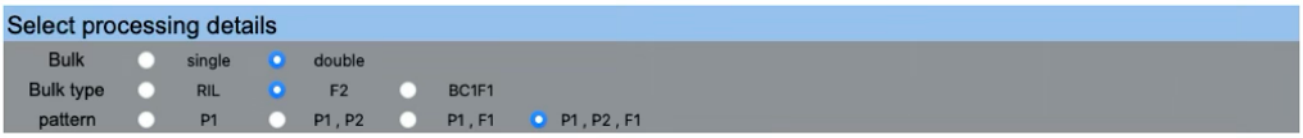
Chromosome name, start position, end position, and peak position are described in tab-delimited format.

①	②	③	④	⑤	⑥	⑦
chr2	0	2050000	1050000	0.76	0.32	0.444
chr03	0	5000000	1300000	0.866	0.183	0.684
chr03	22200000	25000000	23500000	0.858	0.344	0.515
chr08	0	5100000	1800000	0.877	0.252	0.625

①; chr
②; start position
③; end position
④; peak position
⑤; Abulk SNP-index
⑥; Bbulk SNP-index
⑦; delta SNP-index

result

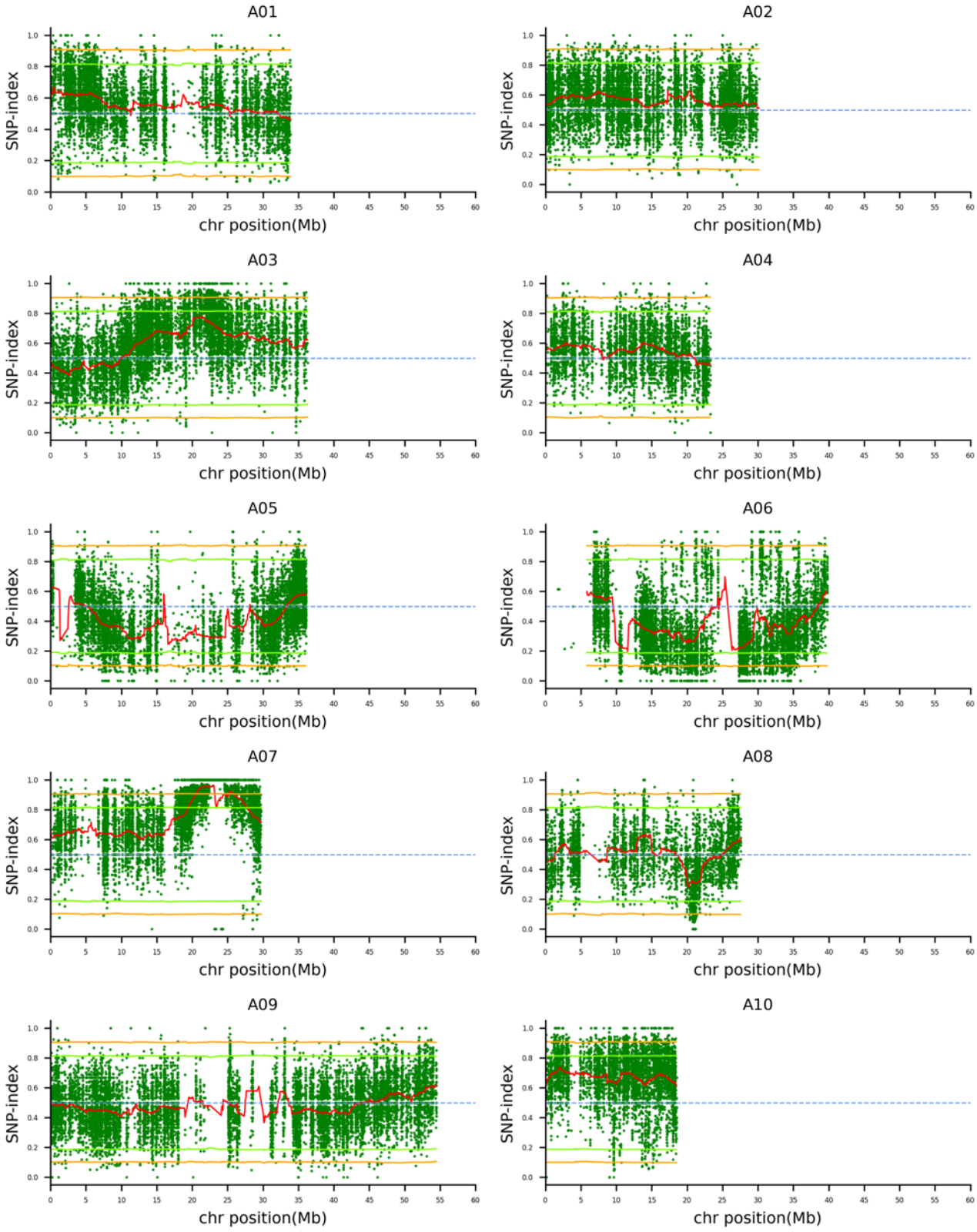
Example of output file when the following contents are selected




```
result
 |---Abulk.png
 |---Bbulk.png
 |---Abulk-Bbulk.png
```

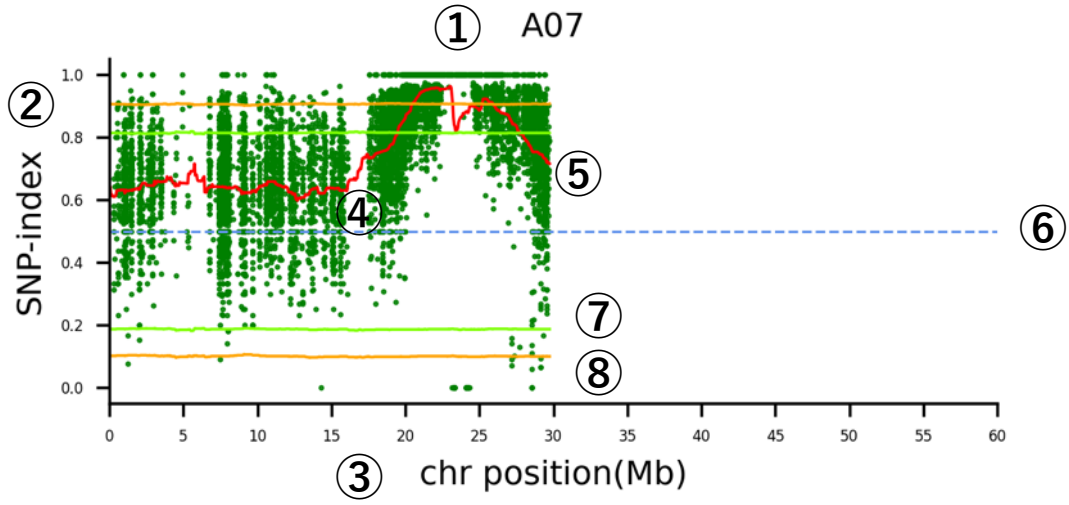
result

Abulk.png



result

Abulk.png



① : Chromosome name

② : SNP-index (0.0 < SNP-index < 1.0)

③ : position (Mb)

④ : SNP
Represents SNPs for position

⑤ : Average SNP-index
Average SNP-index value for Window size

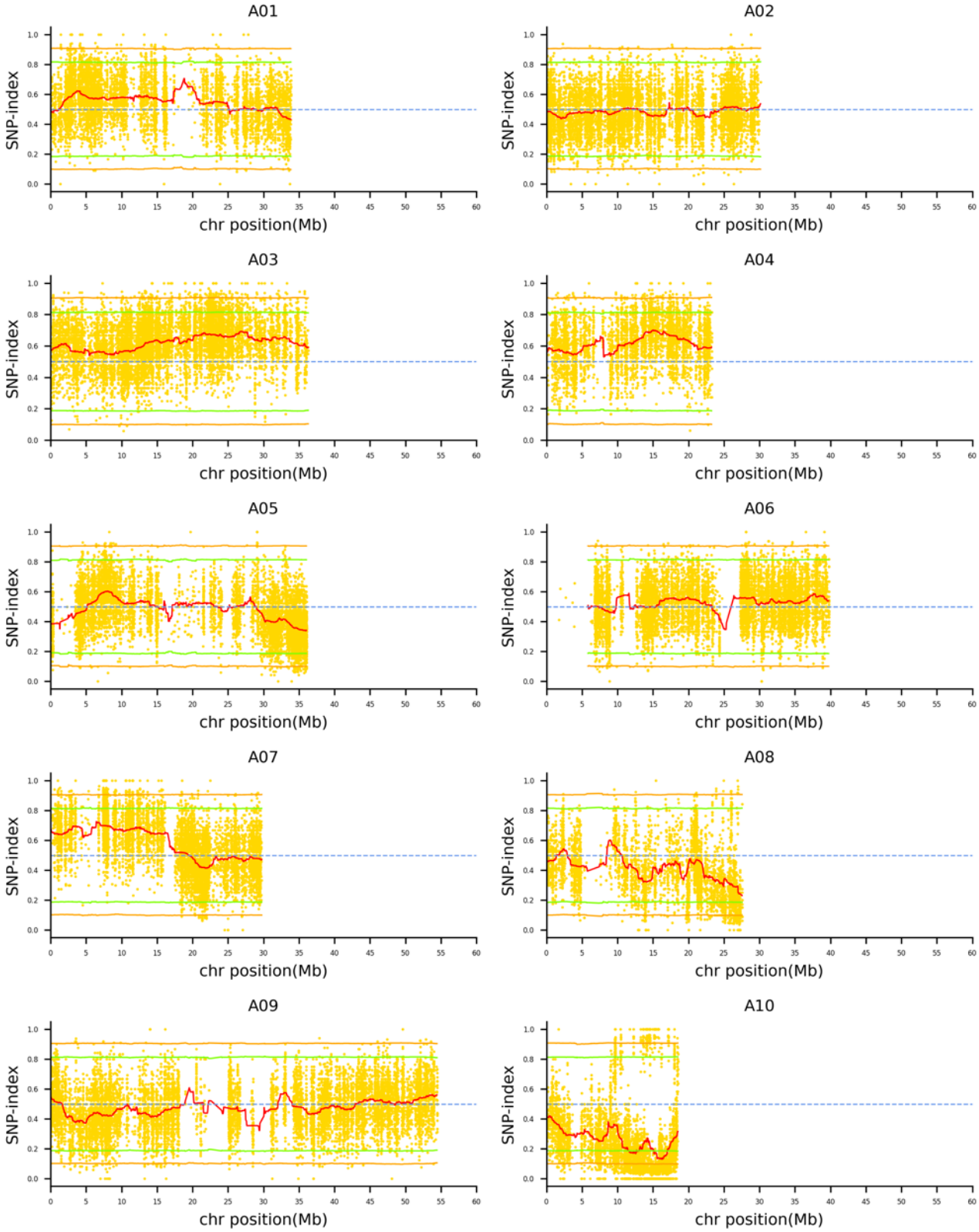
⑥ : Reference line at SNP-index = 0.5

⑦ : 95% confidence interval (depth-based simulation)

⑧ : 99% confidence interval (depth-based simulation)

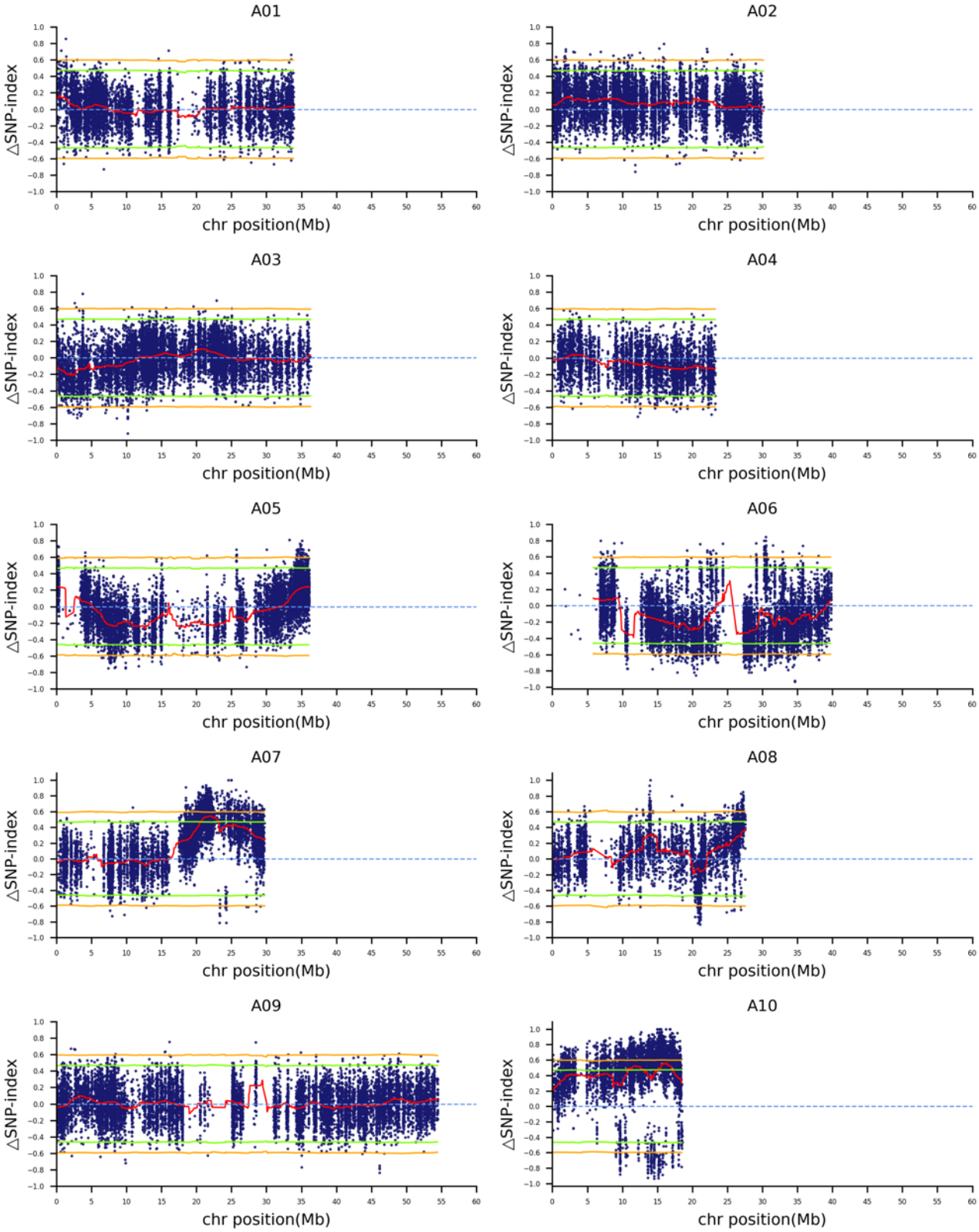
result

Bbulk.png



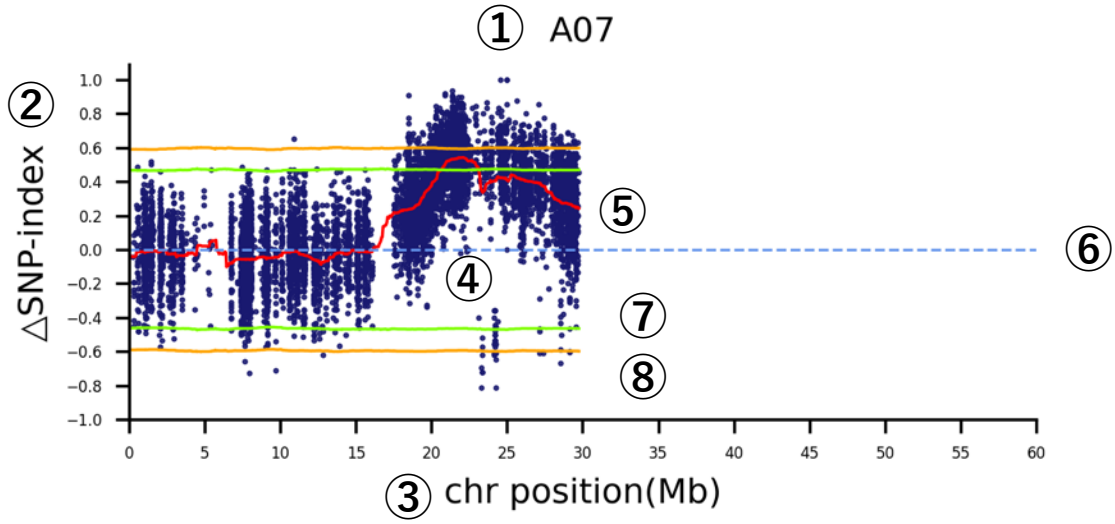
result

Abulk - Bbulk.png



result

Abulk - Bbulk.png



① : Chromosome name

② : SNP-index (-1.0 < SNP-index < 1.0)

③ : position (Mb)

④ : SNP
Represents SNPs for position

⑤ : Average SNP-index
Average SNP-index value for Window size

⑥ : Reference line at SNP-index = 0.0

⑦ : 95% confidence interval (depth-based simulation)

⑧ : 99% confidence interval (depth-based simulation)